Notes: Mann (JFE, 2018)

Creditor Rights and Innovation: Evidence from Patent Collateral

Contents

1 Big picture

- **Question.** Do stronger creditor rights to patents relax financing constraints for innovative firms, or do they instead discourage risk-taking by giving creditors more power in distress?
- **Core idea.** Patents are unusual intangible assets because they can be pledged as collateral. If creditor rights to patent collateral strengthen, then lenders should be more willing to extend debt against existing patent portfolios.
- **Setting.** U.S. patenting firms, especially public Compustat firms, from the late 1970s through 2013. The key legal setting is Delaware's pro-creditor law together with four federal court decisions in 2002, 2003, 2007, and 2009 that increased the relevance of state law for patent ownership and foreclosure.
- **Main empirical design.** Difference-in-differences comparing Delaware-incorporated firms to non-Delaware firms around those four court decisions.
- **Main result.** When creditor rights to patents strengthen, treated firms raise more debt, spend more on R&D, later produce more patents, and become more likely to pledge patents as collateral.
- **Quantitative punchline.** For all firms, debt rises by about 1 percentage point of assets and quarterly R&D rises by about 0.17 percentage points of assets after a decision. In high-tech industries, the effects are larger: debt rises by about 1.45 percentage points of assets and quarterly R&D by about 0.39 percentage points of assets.
- **Mechanism.** The paper's interpretation is collateral value, not generic pro-business Delaware effects. The legal shocks concern patents specifically; placebo tests with trademarks do not show the same pattern.
- **Why it matters.** The paper shows that at least some intangible assets are economically meaningful collateral and that legal clarity over creditor rights can materially affect innovation financing.

2 Data and measurement (key objects)

2.1 Patent collateral data

- **Source.** The paper uses the Patent Assignment Dataset of the USPTO, which records patent ownership transfers and notices of security interests.
- **Key object.** A patent is treated as pledged when the USPTO filing records that it has been used as collateral for a loan.
- **Content of each pledge record.** Date, patent number, borrower name, and lender name.
- **Matching.** Patent collateral records are matched to the NBER patent-data project firm identifier `pdpass`, and then to Compustat `gvkey` for financial analysis.

2.2 Patent, application, and citation data

- **Baseline patent source.** The NBER patent data provide patent grants, citations, classifications, and firm identifiers.

- **Extension.** Because the NBER vintage ends in 2006, Mann extends patent grants and applications through 2013 using Google Patents / USPTO bulk data.
- **Citation measure.** The paper uses citations received within the first five years of a patent's life to limit right-truncation bias.
- **Patent quality proxy.** Citation counts and Hall-Jaffe-Trajtenberg style generality are used as indicators of redeployability and collateral value.

2.3 Compustat panel and financial variables

- **Unit of observation.** A firm-quarter panel after merging patent data to quarterly Compustat.
- **Debt.** Total debt is long-term debt plus short-term debt.
- **Innovation input.** Quarterly R&D expense, cleaned to remove acquired in-process R&D.
- **Scaling.** Financial outcomes are divided by total assets and winsorized at 0 and 1.
- **Sample restriction.** The analysis keeps U.S.-incorporated firms and drops firms with missing or nonpositive assets.

2.4 Main variables used throughout the paper

- **Pledge indicator.** Whether a patent, or a firm's patent portfolio, has ever been pledged as collateral.
- **Redeployability proxies.** Patent citations and generality.
- **Innovation output.** Successful patent applications and citation-weighted patent output.
- **Treatment.** Indicator for Delaware incorporation.
- **Post.** Indicator for quarters after each court decision in the stacked event windows.
- **High-tech industries.** The top ten three-digit SIC industries most represented among patent-collateral borrowers. These are low-tangibility, high-patenting industries.

2.5 Useful sample facts

- In 2013, firms that had ever pledged patents as collateral accounted for about 20% of Compustat R&D and 20% of Compustat patenting.
- In the natural-experiment sample there are 11,212 firms, and about 65% are Delaware-incorporated and therefore treated.
- In the treated-versus-untreated comparison around 2000, Delaware firms do not look different in size or tangibility, but they do have somewhat higher debt, higher R&D intensity, and a slightly higher probability of having pledged patents.

3 Models and empirical specifications (all equations + interpretation)

3.1 Patent-level pledge regression

A first descriptive regression studies which patents are most likely to be pledged:

$$\text{Pledge}_p = \alpha + \beta_1 \text{CitedPatent}_p + \beta_2 \ln(\text{Citations}_p) + \beta_3 \text{Generality}_p + \beta_4 \text{PublicFirm}_p + f(\text{GrantYear}_p) + \varepsilon_p. \tag{1}$$

- **Dependent variable.** Indicator that patent p is ever pledged as collateral.
- **Interpretation.** This is not the causal design; it is a descriptive test of which patents look collateralizable.
- **Economic meaning.** If lenders value redeployability, highly cited and more general patents should be more likely to be pledged.
- **Main result.** They are. A one-standard-deviation increase in generality raises the pledge probability by about 1.2 percentage points, and highly cited patents are also much more likely to be pledged.

3.2 Firm-level pledge regression

The paper then moves to the firm as the relevant decision-making unit:

$$\text{Pledge}_i = \alpha + \beta_1 \ln(\text{Patents}_i) + \beta_2 \ln(\text{Citations/Patent}_i) + \beta_3 \ln(\text{Patents}_i) \times \ln(\text{Citations/Patent}_i) + \Gamma X_i + \varepsilon_i. \tag{2}$$

- **Interpretation.** Firms with larger and higher-quality patent portfolios should find patent collateral more useful.
- **Controls.** Log assets, tangibility, cash, profits, market-to-book, and dividend-payer status.
- **Main result.** Firms with more patents, more highly cited patents, lower cash, and less payout activity are more likely to have pledged patents.
- **Reading the interaction.** The negative interaction says the marginal predictive power of extra citations is strongest when the portfolio is still relatively small.

3.3 Cross-sectional debt and R&D regressions

To show that patent collateral is economically large, the paper relates pledge status to financing and investment:

$$Y_i = \alpha + \beta \, \text{PledgedPatents}_i + \Gamma X_i + \varepsilon_i, \tag{3}$$

where Y_i is either debt/assets or quarterly R&D/assets.

- **Interpretation.** This is descriptive evidence that firms using patent collateral are not doing so on the margin for tiny loans.
- **Main result.** Pledging firms have substantially higher debt ratios, and among high-tech firms they also have significantly higher R&D intensity.
- **Why it matters.** The paper wants to establish that patent collateral is quantitatively important before turning to causal evidence.

3.4 Within-firm dynamics around a patent-backed loan

For firms in the high-tech industries, Mann estimates a distributed-lag event-study around the quarter in which patents are pledged:

$$y_{iq} = \alpha_i + \sum_{\tau=-8}^{8} \beta_\tau \, \text{Pledge}_{i,q-\tau} + \varepsilon_{iq}. \tag{4}$$

- **Outcomes.** Debt issuance, debt retirement, total debt/assets, R&D/assets, and capital expenditures/assets.
- **Interpretation.** This shows the time profile of financing and investment around actual patent-collateral use.
- **Main result.** Total debt jumps by about 4% of assets in the pledge quarter, while R&D rises before and after the deal and stays elevated for roughly a year.
- **Economic meaning.** Patent-backed borrowing appears tightly linked to innovation finance rather than to general capital expenditure.

3.5 Main stacked difference-in-differences specification

The core natural-experiment specification is the paper's Equation (1):

$$y_{iskq} = \alpha \, 1\{DE\}_{sk} + \delta \, 1\{after\}_{kq} + \beta \, 1\{DE\}_{sk} \times 1\{after\}_{kq} + \varepsilon_{iskq}. \tag{5}$$

- **Indices.** i indexes firms, s states of incorporation, $k \in \{1, 2, 3, 4\}$ the four court decisions, and $q \in [-8, 8]$ the quarter in event time within each stacked 16-quarter window.
- **Treated group.** Delaware-incorporated firms.
- **Post indicator.** Quarters after the relevant court decision in each stacked event.
- **Key parameter.** β is the causal effect of stronger creditor rights to patents under the parallel-trends assumption.
- **Outcomes.** Debt/assets, R&D/assets, and later patenting outcomes.
- **Why Delaware matters.** Delaware's ABSFA gives stronger de facto creditor rights if state law governs patent ownership and foreclosure; the court decisions increased exactly that relevance of state law.

3.6 Event-time version of the stacked design

To inspect pretrends and timing, the paper replaces the simple post indicator with event-quarter dummies:

$$y_{iskq} = \alpha\, 1\{DE\}_{sk} + \sum_{\tau \neq 0} \beta_\tau\, 1\{DE\}_{sk} \times 1\{q = \tau\} + \text{FE} + \varepsilon_{iskq}. \tag{6}$$

- **Interpretation.** This is the event-study version of the main DiD.
- **Main result.** For high-tech firms, there is no obvious differential pretrend before the court dates; the effects appear after the decisions.

3.7 Long-run patent-output specification

Because patenting responds slowly, the paper tracks treated and untreated cohorts over time using firm and year fixed effects:

$$\log(\text{CumPatents}_{it}) = \alpha_i + \lambda_t + \sum_{\tau} \beta_\tau\, \text{Delaware}_i \times 1\{t = \tau\} + \varepsilon_{it}. \tag{7}$$

- **Outcome.** The cumulative number of successful patent applications, timed by application year.
- **Interpretation.** The yearly interaction coefficients show how Delaware firms' patenting evolves relative to non-Delaware firms after 2001.
- **Main result.** The patenting effect emerges with a lag and reaches about 19% by 2013.

3.8 Citation-weighted patent output and collateral-use dynamics

The same cohort-based specification is used with two alternative outcomes:

$$\log(\text{CumCitations}_{it}) = \alpha_i + \lambda_t + \sum_{\tau} \beta_\tau\, \text{Delaware}_i \times 1\{t = \tau\} + \varepsilon_{it}, \tag{8}$$

and

$$\text{EverPledged}_{it} = \alpha_i + \lambda_t + \sum_{\tau} \beta_\tau\, \text{Delaware}_i \times 1\{t = \tau\} + \varepsilon_{it}. \tag{9}$$

- **Citation-weighted patents.** This checks whether later patenting growth is just relabeling weak ideas as patents; the effect remains positive and reaches about 10% for citation-weighted output.
- **Ever pledged indicator.** This shows whether treated firms increasingly enter the patent-backed loan market after the legal shocks; by 2013 the probability rises by about 4 percentage points more for treated firms.

4 Identification and instruments (what makes the estimates credible)

4.1 Baseline identification idea

- **Cross-sectional variation.** Delaware has a uniquely pro-creditor law, the ABSFA, governing true-sale treatment of collateral transferred to a bankruptcy-remote SPE.
- **Time-series variation.** Four federal court decisions between 2002 and 2009 clarify that patent ownership and security interests are largely matters of state law rather than exclusively federal patent law.
- **Difference-in-differences logic.** When those decisions raise the importance of state law for patents, Delaware-incorporated patenting firms experience a larger improvement in creditor rights than comparable firms elsewhere.
- **What is being isolated.** The court decisions only concern patent law, so the design isolates changes in patent collateral value rather than broad changes in bankruptcy or contract law.

4.2 Why Delaware's law matters

- **The ABSFA.** Effective January 2002, the law allows collateral to be transferred to a special-purpose entity in a way that protects "true sale" treatment and helps avoid recharacterization in bankruptcy.
- **Economic implication.** If patent collateral can be more cleanly isolated from the borrower's bankruptcy estate, lenders can expect faster and more secure recovery in default.
- **Why this is useful empirically.** State secured-lending law is mostly harmonized across states; Delaware's ABSFA is the main exceptional pro-creditor rule. That makes Delaware a sharp treatment state when patents are found to be governed by state law.

4.3 The four court decisions

- **Rhone-Poulenc Agro v. DeKalb Genetics (March 26, 2002).** Narrows the special federal bona fide purchaser interpretation for patents and points back toward state law.
- **Pasteurized Eggs v. Bon Dente (May 30, 2003).** Holds that state filing procedures are sufficient to perfect a security interest in patents; a separate USPTO filing is not required.
- **Braunstein v. Gateway Management (May 15, 2007).** Holds that state-law procedures are necessary and that relying on USPTO filings alone is insufficient for foreclosure/perfection.
- **Sky Technologies v. SAP (August 20, 2009).** Holds that foreclosure under state law successfully transfers ownership of patents.
- **Common thread.** All four decisions move patents closer to the legal treatment of ordinary assets under state secured-transactions law.

4.4 Why exclusion is plausible

- **Patent specificity.** The decisions interpret patent law, not trademark law, copyright law, or generic bankruptcy policy.
- **Placebo logic.** If the results were really just "Delaware is special," then trademarks should respond too. Table 10 says they do not.
- **Within-firm patterns.** Firms without patents yet, or with weak patent portfolios, do not show the same response. Firms with prior patent collateral use or highly cited portfolios respond the most.

4.5 Main threats and how the paper handles them

- **Nonrandom Delaware incorporation.** Firms choose Delaware, so the strongest causal claim is closer to an ATT than a universal ATE. The paper is explicit about this.
- **Parallel trends.** Event-time plots for debt and R&D are flat before treatment, and placebo years without court decisions show no pattern.
- **Regional confounds.** Results are robust to headquarters-state fixed effects and hold in California, the Northeast, Texas/Colorado, and Florida/Georgia.
- **Other states with similar laws.** Firms incorporated in states with similar legislation are dropped.

5 Tables and figures

5.1 Figure 1: Growth in the market for patent-backed loans

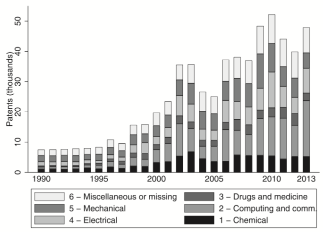
Read:

- Patent collateral use rises sharply over time, both in levels and as a share of patent vintages.
- By recent years in the sample, more than 40,000 patents per year are pledged, and about 15% of a vintage is pledged within five years.
- This matters because it shows patent collateral is not a niche curiosity; it is a growing financing channel.

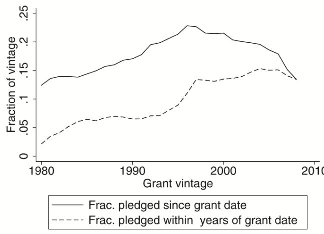
5.2 Table 1: Which patents are pledged?

Read:

- Highly cited patents and patents with greater generality are more likely to be pledged.
- The evidence fits a collateral-value story: patents with broader redeployability are more useful to lenders.
- Public-company patents are less likely to be pledged, consistent with easier access to equity or other finance.



A. Number of US utility patents pledged as collateral per year, 1990–2013, as recorded in the Patent Assignment Dataset of the United States Trademark and Patent Office (USPTO). The six patenting categories are taken from Hall, Jaffe, and Trajtenberg (2001).



B. The solid line shows the fraction of each vintage of utility patents granted to US companies that have been pledged as collateral. The series begins to decrease before 2000, but this is largely an artifact of right truncation, since the younger vintages have had less time to be pledged as collateral. To illustrate this, the dotted line shows the fraction of patents that are pledged within five years of their grant date.

Fig. 1. Growth in the market for patent-backed loans over time.

Table 1
Predictive regressions of the decision to pledge patents, patent level.

The sample is all utility patents granted to US corporations through 2013, as recorded in the Patent Grant Dataset of the United States Patent and Trademark Office (USPTO), that can be matched to a *pdpass* identifier from the Patent Data Project of the National Bureau of Economic Research. The outcome variable is an indicator for the patent ever being pledged as collateral in the Patent Assignment Database of the USPTO. The "Cited patent" indicator and the log citation count are based on the first five years of the patent's term, where citations are extracted from patent grant data. The Generality measure of each patent is computed as in Hall et al. (2001), and is demeaned and scaled by its sample standard deviation. "Public company" is an indicator for whether the *pdpass* identifier can be further matched to a Compustat company. Standard errors are clustered by the 36 two-digit technology subcategories introduced in Hall et al. (2001).

	(1) Pledge	(2) Pledge	(3) Pledge	(4) Pledge	(5) Pledge
Cited patent	0.0269***				
	(0.00553)				
Ln(citations)		0.0151***		0.0141***	0.00733**
		(0.00315)		(0.00307)	(0.00273)
Generality			0.0124***	0.00867***	0.00587*
			(0.00258)	(0.00256)	(0.00293)
Public company				−0.0672***	−0.0648***
				(0.00699)	(0.00777)
1996–Grant year					0.0000391
					(0.000479)
$(1996\text{–Grant year})^2$					−0.000336***
					(0.0000315)
Constant	0.155***	0.164***	0.178***	0.204***	0.240***
	(0.00907)	(0.00784)	(0.00699)	(0.00651)	(0.00814)
Obs.	1389533	1193799	1389533	1193799	1193799
R^2	0.000597	0.00141	0.00104	0.00919	0.0155

Standard errors in parentheses
* $p < 0.10$
** $p < 0.05$
*** $p < 0.01$

5.3 Table 2: Which firms pledge patents?

Read:

- Firms with bigger patent stocks and more highly cited portfolios are more likely to use patent collateral.
- Financially constrained firms are also more likely to pledge: they hold less cash and are less likely to pay dividends.
- The interaction says patent quality matters most when the portfolio is not already very large.

5.4 Figures 2 and 3: Where patent collateral is used

Read:

- The main borrowing industries are classic high-tech, low-tangibility sectors such as medical instruments, electronic components, data processing, drugs, communications equipment, and lab instruments.
- Firms that have ever pledged patents account for a large share of aggregate innovative activity: around 20% of Compustat R&D and patenting by 2013.
- This supports the paper's claim that patent collateral matters exactly where tangible collateral is scarce.

5.5 Table 3 and Figure 4: How large are patent-backed loans?

Read:

- Table 3 shows that patent-pledging firms carry more debt, and in high-tech sectors they also spend more on R&D.

Table 2
Predictive regressions of the decision to pledge patents, firm level.

The sample is the cross-section of Compustat firms in 2010q1 that had produced at least one patent by that date, using the sample of patents from Table 1, and that have non-missing data for each of the explanatory variables in this table. The outcome variable is an indicator for whether the firm had ever pledged patents as collateral in the USPTO Assignment Database as of 2010q1. Ln(Patents) is the total number of patents the firm had produced by 2010q1, and Ln(Citations/Patent) is the log average number of citations received by those patents. Citation counts are based on the first five years of the patent's term. The financial variables from Compustat are averaged over the prior ten years. "Dividend payer" is an indicator for the firm having a positive average dividend payout over the prior ten years. Standard errors are clustered by three-digit SIC.

	(1) Pledge	(2) Pledge	(3) Pledge	(4) Pledge
Ln(Patents)	0.0704***		0.0825***	0.0707***
	(0.0137)		(0.0124)	(0.0143)
Ln(Citations/Patents)	0.0446***		0.0511***	0.0539***
	(0.0160)		(0.0173)	(0.0158)
Ln(Patents) × Ln(Citations/Patents)	−0.0205***		−0.0189***	−0.0146**
	(0.00785)		(0.00644)	(0.00724)
Ln(Assets)		−0.00381	−0.0301***	−0.0312***
		(0.00482)	(0.00506)	(0.00558)
PP&E/Assets		−0.196**	−0.126*	−0.0788
		(0.0772)	(0.0666)	(0.106)
Cash/Assets		−0.513***	−0.565***	−0.638***
		(0.0536)	(0.0606)	(0.0648)
Profits/Assets		0.0463	0.0654**	0.0619*
		(0.0335)	(0.0327)	(0.0329)
Market/Book		−0.0000237	−0.00000608	−0.0000203
		(0.0000630)	(0.0000624)	(0.0000595)
Dividend payer		−0.0868***	−0.0871***	−0.0753***
		(0.0267)	(0.0255)	(0.0265)
Constant	0.0878***	0.519***	0.424***	0.450***
	(0.0254)	(0.0416)	(0.0462)	(0.0460)
Fixed effect	None	None	None	3-digit SIC
Obs.	1653	1653	1653	1653
R^2	0.0434	0.0521	0.110	0.0912

Standard errors in parentheses
* $p < 0.10$
** $p < 0.05$
*** $p < 0.01$

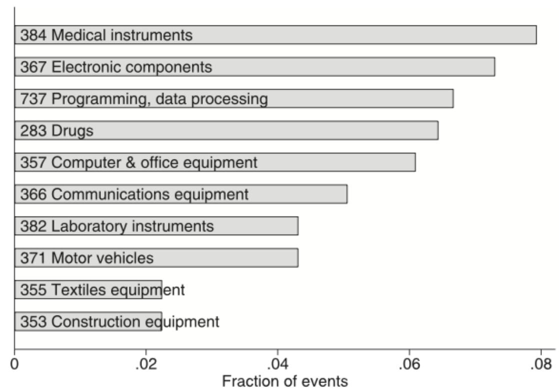


Fig. 2. Top ten standard industry classifications of borrowers against patent collateral, by number of deals. The sample is as in Table 1, but restricted to patent pledge events for which the patent owner can be matched to Compustat, and retaining only one observation per firm-quarter. The unit of observation is thus a firm-level patent pledge event.

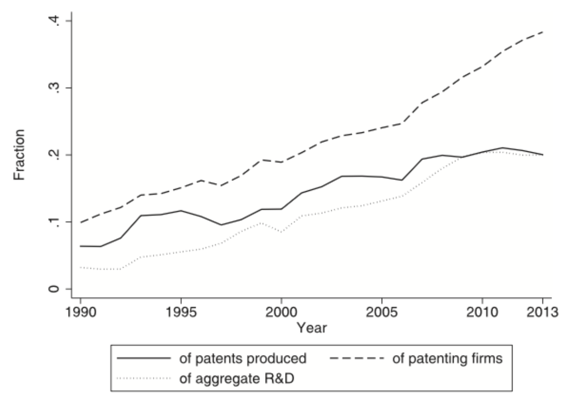


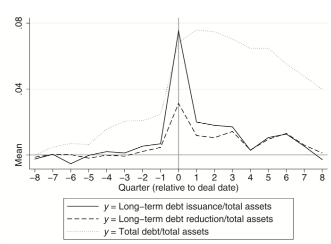
Fig. 3. Fraction of Compustat patenting and R&D expense performed by companies that have pledged their patents as collateral at some point in the past. Each year, companies in Compustat are divided according to whether or not they can be matched to a patent pledge event, in that or any prior year, in the USPTO Patent Assignment Dataset. The figure plots the fraction of each year's patents produced (using the sample from Table 1), of patenting companies, and of aggregate Compustat R&D attributable to those companies.

Table 3
Cross-sectional association between the decision to pledge patents, debt usage, and R&D investment.

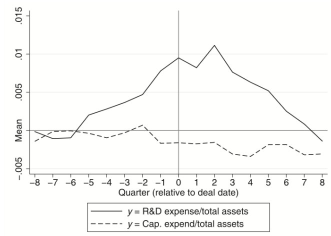
The sample is the same as in Table 2: the cross-section of Compustat firms from 2010q1 that had produced at least one patent by that date and that have nonmissing data for each of the explanatory variables. The outcome variable in the first three columns is the ratio of total debt to total assets. In the last column, it is the ratio of quarterly R&D expense to total assets. The first explanatory variable is an indicator for whether the firm had pledged patents as collateral in the USPTO Assignment Dataset by 2010q1 (the same as the outcome variable in Table 2). The financial variables from Compustat are averaged over the prior ten years. "Dividend payer" is an indicator for the firm having a positive average dividend payout over the prior ten years. Standard errors in columns 1 and 2 are clustered by three-digit SIC code. Columns 3 and 4 restrict to the ten high-tech SIC codes identified in Fig. 2. In these columns, standard errors are clustered by the company's state of location.

	(1) Debt/Assets	(2) Debt/Assets	(3) Debt/Assets	(4) R&D/Assets
Pledged patents	0.0647***	0.0435***	0.0278**	0.00505**
	(0.00877)	(0.00903)	(0.0103)	(0.00239)
Ln(Assets)	0.00775***	0.00145	−0.00428	−0.00627***
	(0.00282)	(0.00298)	(0.00327)	(0.00120)
PP&E/Assets	0.324***	0.197***	0.216***	0.0220
	(0.0468)	(0.0372)	(0.0633)	(0.0149)
Market/Book	0.0000363	0.0000967	0.000258	0.0000518
	(0.0000543)	(0.0000960)	(0.000242)	(0.0000343)
Profits/Assets	−0.157***	−0.153***	−0.131***	−0.0729***
	(0.0237)	(0.0249)	(0.0320)	(0.0213)
Cash/Assets		−0.233***	−0.187***	0.0942***
		(0.0657)	(0.0291)	(0.00811)
Dividend payer		−0.00341	−0.00764	−0.00348
		(0.00887)	(0.0128)	(0.00260)
Constant	0.0618***	0.192***	0.200***	0.0361***
	(0.0163)	(0.0230)	(0.0320)	(0.00749)
Sample	All	All	High-tech	High-tech
Obs.	1653	1653	889	889
R^2	0.199	0.259	0.221	0.546

Standard errors in parentheses
*$p < 0.10$
** $p < 0.05$
*** $p < 0.01$



A. Financing activity around dates of patents being pledged as collateral. The outcome variables are quarterly debt issuance (Compustat item DLTISQ; solid line), debt reduction (item DLTR; dashed line), and total debt (DLTTQ + DLTQ; dotted line), all divided by total assets (ATQ).



B. Investment activity. The outcome variables are quarterly R&D (solid line) and capital expenditures (dashed line), both divided by total assets. See Section 2 for details on R&D measurement.

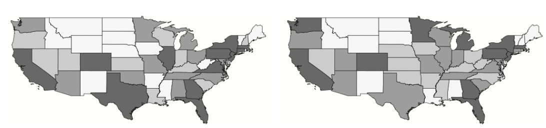
Fig. 4. Dynamics of financing and investment for high-tech companies around dates of patents being pledged as collateral. The sample is quarterly Compustat data, 1990–2013, restricting to companies that can be matched with a patent pledge event in the USPTO Assignment Dataset, and also restricting to the ten three-digit SIC industries identified in Fig. 2. The figure plots the coefficients β_τ from the specification $y_{iq} = \alpha_i + \sum_{\tau=-8}^{8} \beta_\tau Pledge_{q-\tau} + \epsilon_{iq}$, where i indexes firms.

- Figure 4 shows the within-firm dynamics: debt jumps by about 4% of assets in the pledge quarter.
- R&D rises around the same time, while capital expenditures do not. That is crucial: the financing appears to fund intangible investment rather than ordinary capex.

5.6 Table 4 and Figure 5: The natural experiment setup

Table 4
Court decisions used in the natural experiment.

Case	Decision date
Rhone-Poulenc Agro v DeKalb Genetics Corp.	March 26, 2002
Pasteurized Eggs Corporation v Bon Dente Joint Venture	May 30, 2003
Braunstein v Gateway Management Services	May 15, 2007
Sky Technologies LLC v SAP AG and SAP America	August 20, 2009



A. Treated firms. **B.** Control firms.

Fig. 5. Geographic distribution of sample firms by headquarters state. The sample is the set of firms used in the natural experiment, as defined in Table 5. States are shaded according to quartiles of the number of sample firms headquartered there.

Read:

- Table 4 lists the four court decisions that create the time-series component of the design.
- Figure 5 shows treated and control firms are geographically dispersed across the United States rather than concentrated in a single region.
- Together they motivate the design: Delaware law creates cross-sectional variation; the decisions create timing variation.

5.7 Table 5: Treated versus untreated firms before treatment

Read:

- Treated and untreated firms look similar in assets, tangibility, and patent counts.
- Delaware firms do have somewhat higher debt and R&D intensity.
- The table therefore highlights the need for a difference-in-differences design rather than a simple cross-sectional comparison.

5.8 Tables 6 and 7: Main causal effects on debt and R&D

Table 6

Increase in debt due to treatment, robustness to fixed effects, and stronger effects for high-tech firms.

The sample is all Compustat company-quarters within eight quarters of the four decision dates listed in Table 4, stacked into one event-time dataset. *Treated* is an indicator for being incorporated in Delaware, and *After* is an indicator for being after the relevant decision date. All fixed effects listed at the bottom of the table are interacted with dummies for the individual decisions. *HQ* is the company's state of headquarters, and *Inc.* is its state of incorporation, both taken from Compustat. High-tech industries are those listed in Fig. 2. Standard errors are clustered at the level of the fixed effect included in the regression, or by state of incorporation when no fixed effect is included.

	(1) Debt/Assets	(2) Debt/Assets	(3) Debt/Assets	(4) Debt/Assets	(5) Debt/Assets
Treated	0.0204**	0.0249***	−0.0125	−0.0113	
	(0.00974)	(0.00669)	(0.0141)	(0.00782)	
After	−0.0131***	−0.0132***	−0.00848*	−0.00887*	−0.00411
	(0.00218)	(0.00230)	(0.00464)	(0.00522)	(0.00471)
Treated, after	0.0102***	0.0100***	0.0145***	0.0148***	0.0103*
	(0.00218)	(0.00264)	(0.00464)	(0.00426)	(0.00559)
Fixed effect	None	HQ	None	HQ	HQ x Inc.
Sample	All	All	High-tech	High-tech	High-tech
Obs.	412246	412246	124136	124136	124136
R^2	0.00225	0.00273	0.000227	0.000198	0.000121

Standard errors in parentheses
* $p < 0.10$
** $p < 0.05$
*** $p < 0.01$

Table 7

Increase in R&D due to treatment, robustness to fixed effects, and stronger effects for high-tech firms.

The sample is all Compustat company-quarters within eight quarters of the four decision dates listed in Table 4, stacked into one event-time dataset. *Treated* is an indicator for being incorporated in Delaware, and *After* is an indicator for being after the relevant decision date. All fixed effects listed at the bottom of the table are interacted with dummies for the individual decisions. *HQ* is the company's state of headquarters, and *Inc.* is its state of incorporation, both taken from Compustat. High-tech industries are those listed in Fig. 2. Standard errors are clustered at the level of the fixed effect included in the regression, or by state of incorporation when no fixed effect is included.

	(1) R&D/Assets	(2) R&D/Assets	(3) R&D/Assets	(4) R&D/Assets	(5) R&D/Assets
Treated	0.00948***	0.00859***	0.0140***	0.0107***	
	(0.00210)	(0.00105)	(0.00392)	(0.00257)	
After	−0.00115***	−0.000924**	−0.00171*	−0.00152	−0.000620
	(0.000378)	(0.000400)	(0.000851)	(0.00120)	(0.00115)
Treated, after	0.00169***	0.00157***	0.00394***	0.00371***	0.00285*
	(0.000378)	(0.000478)	(0.000851)	(0.00120)	(0.00151)
Fixed effect	None	HQ	None	HQ	HQ x Inc.
Sample	All	All	High-tech	High-tech	High-tech
Obs.	412246	412246	124136	124136	124136
R^2	0.00580	0.00433	0.00580	0.00322	0.000114

Standard errors in parentheses
* $p < 0.10$
** $p < 0.05$
*** $p < 0.01$

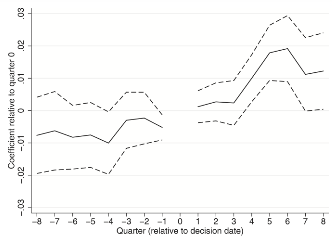
Read:

- Table 6 is the core debt result: treated firms increase debt/assets by about 0.010 in the full sample and about 0.015 in high-tech industries.
- Table 7 is the parallel R&D result: quarterly R&D/assets rises by about 0.0017 in the full sample and about 0.0039 in high-tech industries.
- The robustness to headquarters-state fixed effects and even headquarters-by-incorporation fixed effects makes the result much more credible.
- The implied pass-through from marginal debt to annual R&D is economically meaningful, especially in high-tech sectors.

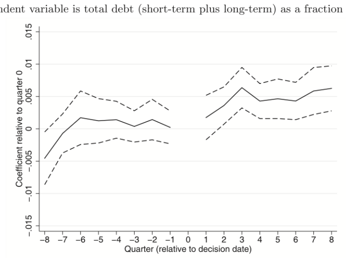
5.9 Figure 6: Event-study evidence

Read:

- Pretrends are basically flat for both debt and R&D in the high-tech sample.
- Effects appear shortly after the court decisions rather than drifting in beforehand.
- This is the main visual support for the parallel-trends assumption.



A. The dependent variable is total debt (short-term plus long-term) as a fraction of total assets.



B. The dependent variable is quarterly R&D expense as a fraction of total assets.

Fig. 6. Regression coefficients in event time. The analysis is identical to column 3 of Tables 6 and 7, except that the *After* indicator is replaced with a full set of event-quarter dummies (with $q = 0$ representing the court decision date). As in those tables, the sample is restricted to the high-tech industries that exhibit the strongest effects, and standard errors are clustered by the firm's state of incorporation. Dashed lines are two standard errors above and below the estimated coefficients.

Table 8
Effects of the natural experiment, separated by proxies for financial constraints.

The sample and specification are as in column 1 of Tables 6 and 7, but adding interactions of all explanatory variables with a binary proxy for greater financial constraints. For reasons of space, only the interaction terms of interest are reported. The financial-constraints proxy used in each column is listed at the bottom of the table.

A. The outcome variable is total debt over total assets.

	(1) Debt/AT	(2) Debt/AT	(3) Debt/AT	(4) Debt/AT	(5) Debt/AT
Treated, after	0.00699***	0.00663***	0.00544	0.00218	0.00287
	(0.00191)	(0.00240)	(0.00332)	(0.00174)	(0.00261)
Treated, after, proxy	0.00409	0.00702**	0.00788*	0.00720*	0.0183***
	(0.00372)	(0.00310)	(0.00425)	(0.00367)	(0.00615)
Constraint proxy	AT < $200m	PP&E < $20m	Tang < 15%	No dividend	Unprofitable
Obs.	412246	398550	398550	412246	412246
R^2	0.0131	0.0252	0.0466	0.00320	0.00671

B. The outcome variable is quarterly R&D expense over total assets.

	(1) R&D/AT	(2) R&D/AT	(3) R&D/AT	(4) R&D/AT	(5) R&D/AT
Treated, after	0.000116**	0.0000738	0.000238	0.0000529	0.000342**
	(0.0000449)	(0.0000894)	(0.000632)	(0.000174)	(0.000140)
Treated, after, proxy	0.00292***	0.00363***	0.00289***	0.00146**	0.00387**
	(0.000844)	(0.000770)	(0.000823)	(0.000671)	(0.00154)
Constraint proxy	AT < $200m	PP&E < $20m	Tang < 15%	No dividend	Unprofitable
Obs.	412246	398550	398550	412246	412246
R^2	0.0540	0.0484	0.0153	0.0257	0.0840

Standard errors in parentheses
* $p < 0.10$
** $p < 0.05$
*** $p < 0.01$

5.10 Table 8: Heterogeneity by financial constraints

Read:

- The debt and R&D effects are stronger for smaller firms in PP&E, firms with lower tangibility, non-dividend payers, and unprofitable firms.
- This is exactly what the financing-constraints interpretation predicts.
- The legal shock matters most where collateral scarcity was likely binding.

5.11 Tables 9 and 10: Heterogeneity by patent portfolio and placebo with trademarks

Read:

- Firms with no patents yet do not respond, which is a clean placebo.
- The debt response is stronger for firms with more patents, for firms that already pledged patents, and for firms with highly cited patent portfolios.
- Trademark portfolios do not generate parallel heterogeneity. That sharply strengthens the claim that the legal shock operates through patents specifically.

5.12 Figure 7: Long-run effect on patent output

Read:

- There is little effect immediately after 2001–2003, but a gradually building positive effect emerges from 2004 onward.
- By 2013, treated firms have roughly 19% more cumulative successful patent applications.
- This lagged pattern fits the idea that extra debt first finances research, which only later becomes observable patent output.

Table 9
Moderating effects of a company's patenting history.

In column 1, the specification is the same as in column 1 of Table 6, while the sample is restricted to companies that are eventually observed to produce patents, but have not done so as of the beginning of the event window. In column 2, the sample is restricted to companies that are eventually observed to produce patents and have produced zero to six patents as of the beginning of the event window, and *# patents* is the count of those patents. In column 3, the sample is the same as in column 2, and *Has pledged* is an indicator for whether the company has been observed to pledge patents as collateral at any point prior to the beginning of the event window. In column 4, the sample is all companies that have produced at least one patent as of the beginning of the event window, and *High cites* is an indicator for whether the average number of citations received by the company's patents in the first five years after being granted is at least four.

	(1) Debt/Assets	(2) Debt/Assets	(3) Debt/Assets	(4) Debt/Assets
Treated	−0.0517	−0.0362	−0.0256	0.0284***
	(0.0366)	(0.0278)	(0.0153)	(0.00948)
After	0.000359	−0.00668	−0.0113**	−0.0176***
	(0.0136)	(0.00779)	(0.00459)	(0.00392)
Treated, after	−0.00141	0.0000861	0.0100**	0.00445
	(0.0136)	(0.00779)	(0.00459)	(0.00392)
# patents		−0.00624		
		(0.00671)		
Treated × # patents		0.00843		
		(0.00671)		
After × # patents		−0.00409		
		(0.00298)		
Treated × After × # patents		0.00604*		
		(0.00298)		
Has pledged			0.0667	
			(0.0427)	
Treated, has pledged			0.0885**	
			(0.0427)	
After, has pledged			−0.0625***	
			(0.0183)	
Treated, after, has pledged			0.0373**	
			(0.0183)	
High cites				−0.0685***
				(0.0137)
Treated, high cites				−0.0320**
				(0.0137)
After, high cites				0.00669
				(0.00543)
Treated, after, high cites				0.0113**
				(0.00543)
Constant	0.239***	0.241***	0.225***	0.238***
	(0.0366)	(0.0278)	(0.0153)	(0.00948)
Sample	No patents yet	0–6 patents	0–6 patents	1+ patents
Obs.	22667	74589	74589	124652
R^2	0.00686	0.00224	0.0144	0.0308

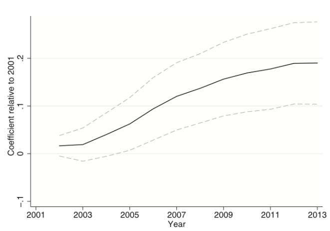
Standard errors in parentheses
* $p < 0.10$
** $p < 0.05$
*** $p < 0.01$

Table 10
Comparison of moderating effects of patents and trademarks.

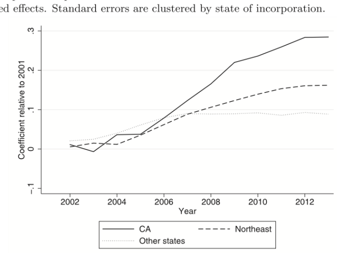
In column 1, the sample is the same as in column 2 of Table 9: patent-producing companies that had six or fewer patents as of the beginning of the event window. In column 2, the sample is trademark-producing companies that had six or fewer trademarks as of the beginning of the event window. In columns 3 and 4, the sample is the intersection of the samples from columns 1 and 2. The regressors *# trademarks* and *# patents* are the number of trademarks registered, and the number of patents granted, as of the beginning of the event window. Standard errors are clustered by state of incorporation.

	(1) Debt/Assets	(2) Debt/Assets	(3) Debt/Assets	(4) Debt/Assets
Treated	−0.0139	−0.0129	−0.118*	−0.0676
	(0.0166)	(0.0124)	(0.0587)	(0.0422)
After	−0.0150***	−0.00946	−0.0336	0.00986
	(0.00474)	(0.00676)	(0.0202)	(0.0163)
Treated, after	0.0110**	0.00397	0.0214	−0.0250
	(0.00474)	(0.00676)	(0.0202)	(0.0163)
# trademarks	0.000501	−0.00806*	−0.0386**	
	(0.000303)	(0.00430)	(0.0145)	
Treated × # trademarks	−0.000599*	−0.00125	0.0305**	
	(0.000303)	(0.00430)	(0.0145)	
After × # trademarks	0.0000234	−0.000533	0.00909	
	(0.000107)	(0.00251)	(0.00544)	
Treated × After × # trademarks	0.0000290	0.00132	−0.00748	
	(0.000107)	(0.00251)	(0.00544)	
# patents				−0.0129
				(0.0117)
Treated × #patents				0.0176
				(0.0117)
After × # patents				−0.0113**
				(0.00481)
Treated × After × #patents				0.0173***
				(0.00481)
Constant	0.224***	0.262***	0.326***	0.246***
	(0.0166)	(0.0124)	(0.0587)	(0.0422)
Sample	0–6 patents	0–6 trademarks	0–6 both	0–6 both
Obs.	74589	90401	22364	22364
R^2	0.00172	0.00384	0.0186	0.00971

Standard errors in parentheses
* $p < 0.10$
** $p < 0.05$
*** $p < 0.01$

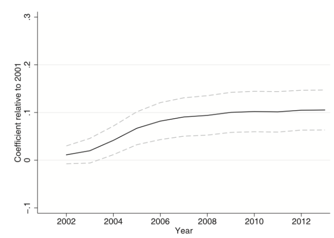


A. Effect on cumulative patent output for treated and untreated firms. Dashed lines are two standard errors above and below the estimated effects. Standard errors are clustered by state of incorporation.

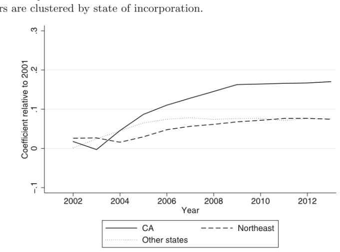


B. Separate effects for firms in California, in the Northeast (NY, MA, CT, RI, NJ, PA), and in the rest of the country.

Fig. 7. Effect on cumulative patent output for treated and untreated firms. The sample is the pooled 2001–2013 panel of all firms in Compustat that were used in the natural experiment summarized in the previous tables, aggregated to an annual frequency by retaining only the last calendar observation for each firm in each year. Using this panel, I estimate a linear model in which the outcome variable is the log of each firm's cumulative count of successful patent applications filed, and the explanatory variables are firm dummies, year dummies, and interactions between the year dummies and treatment status. The figures plot the estimated coefficients on the interaction terms, thus capturing the differential evolution of treated firm's patenting output as the court decisions listed in Table 4 were announced. In Panel A, the estimation is performed on the full sample. In Panel B, the estimation is performed separately across the three major geographic regions in the sample, with companies separated into regions by their state of headquarters.



A. Effect on cumulative citation-weighted patenting output for treated and untreated firms. Citations are measured in the first five years of the patent's term. Dashed lines are two standard errors above and below the estimated effects. Standard errors are clustered by state of incorporation.

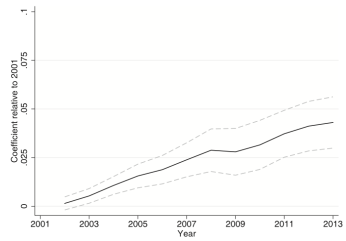


B. Separate effects for firms in California, in the Northeast (NY, MA, CT, RI, NJ, PA), and in the rest of the country.

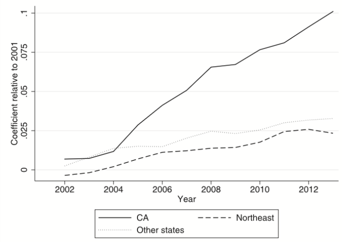
Fig. 8. This figure repeats the analysis of Fig. 7 but weights the cumulative patent counts by the number of citations each patent received in the first five years of its life, or through 2013, whichever is sooner.

5.13 Figure 8: Citation-weighted patent output

Read:



A. Effect on probability of having pledged patents in the past. Dashed lines are two standard errors above and below the estimated effects. Standard errors are clustered by state of incorporation.



B. Separate effects for firms in California, in the Northeast (NY, MA, CT, RI, NJ, PA), and in the rest of the country.

Fig. 9. This figure repeats the analysis of Fig. 7, but the outcome variable is an indicator for whether the firm has ever pledged patents as collateral.

- Citation-weighted patent output also rises, though the effect is smaller than for raw counts and levels off at about 10%.
- This weakens the alternative story that firms are just relabeling low-value ideas as patents.
- The new patenting appears to be at least reasonably valuable, not obviously low quality.

5.14 Figure 9: Later use of patents as collateral

Read:

- Treated firms become increasingly more likely to pledge patents as collateral after the court decisions.
- By 2013, the probability of ever having pledged patents is about 4 percentage points higher for treated firms.
- This closes the loop of the paper: stronger rights raise collateral value, which raises borrowing, which raises innovation, which then expands the stock of pledgeable patents.

6 Short summary

- **Conclusion.** Patent collateral is real, quantitatively important, and sensitive to legal rules governing creditor rights.
- **Causal evidence.** Delaware-incorporated patenting firms increase debt and R&D after court decisions that make state law more relevant to patent ownership and foreclosure.
- **Mechanism.** The strongest effects appear exactly where the collateral channel should be strongest: high-tech firms, financially constrained firms, firms with patents already in hand, firms with highly cited patents, and not firms differentiated by trademarks.

- **Long-run implication.** Better pledgeability of patents feeds into greater patent output and broader later use of patents as collateral.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- **Main finding.** Clarifying creditor rights to patents increases patenting firms' borrowing capacity and innovation investment.
- **Magnitude.** The debt effect is about 1 percentage point of assets in the full sample and 1.45 percentage points in high-tech industries. The quarterly R&D effect is about 0.17 percentage points of assets in the full sample and 0.39 percentage points in high-tech industries.
- **Dynamic implication.** Those input changes later show up in output: treated firms produce more patents and more citation-weighted patents.
- **Mechanism evidence.** The response is strongest for firms with patent-rich and citation-rich portfolios, absent for firms with no patents yet, and absent when replacing patents with trademarks.
- **Interpretation.** The paper reads the legal shocks as increases in patent collateral value, which relax financial constraints rather than suppressing risk-taking.

7.2 Economic intuition

- Patents are intangible, but they are also legally enforceable property rights with identifiable owners and potential buyers. That makes them much more collateralizable than the generic idea of "intangibles" suggests.
- Once courts clarify that state secured-transactions law governs perfection and foreclosure, Delaware's pro-creditor rules become more valuable for patenting firms incorporated there.
- More recoverable collateral raises lenders' willingness to lend. Firms that were constrained can then finance more R&D, and after a lag their innovation output rises.

7.3 Takeaway

This paper is best read as evidence that the legal infrastructure around collateral can shape innovation just as much as the usual factors like taxes, subsidies, or venture capital conditions. The key empirical insight is not merely that patents correlate with borrowing, but that legally stronger creditor rights to patents cause more debt, more research, and more later patenting. In short: when patents become better collateral, innovation becomes easier to finance.